

AirSense Pakistan

Vendor Pricing Report & Budget Appendix

Hardware Procurement - Campus Pilot Phase 1 | Beaconhouse International College

August 2026 Price Review Date	PKR 17,787 Baseline Hardware Subtotal*	PKR 20,000 Total Requested incl. ~5% contingency	3-7 Days Expected Procurement Window
---	--	--	--

* Radiation shield (Item 11) excluded - local quotation pending. Prices are supplier-listed figures and must be reconfirmed at point of order.

1. Vendor Overview

All components are sourced exclusively through domestic vendors and Pakistan-based e-commerce channels to eliminate import delays and customs exposure. The updated sourcing plan replaces the previous Hallroad-centric approach with a set of specialist online electronics suppliers offering authenticated stock, formal listings, and reliable Islamabad delivery.

Vendor / Channel	Type	Est. Delivery	Reliability	Best Used For
Embeded Studio	Online Electronics	2–5 days	HIGH	PMS7003 particulate sensor — primary sensing component
Digilog.pk	Online Electronics	2–5 days	HIGH	ESP32 microcontroller, Rain Sensor A
Electrobes	Online Electronics	2–5 days	HIGH	BME280 environmental sensor, MicroSD module, Rain Sensor B, UNI-T anemometer
A.E Solution	Electrical / Online	2–5 days	HIGH	IP65-rated weatherproof electrical enclosure
Daraz.pk	Marketplace	2–7 days	VERIFY	MicroSD card, jumper wires, cable ties — verify seller ratings and return terms before purchase
Clopal Online	Brand / Online	2–5 days	HIGH	Clopal 10 m heavy-duty power extension lead (baseline option)
Powerhouse Express	Online	2–5 days	HIGH	Camelion CMS-178 extension reel (alternative option — replaces Clopal in that scenario)

Expert Tools World	Online	2–5 days	HIGH	GMSA RTV Silicone weatherproof sealant
Electronics Hub	Online Electronics	2–5 days	HIGH	5V 2A AC/DC power adapter
DreamsMart.pk	Online	2–5 days	HIGH	Stainless-steel 2-inch hose clamps
Multan Electronics	Online / Physical	2–5 days	HIGH	M3–M6 mild-steel nuts and bolts — confirm final sizes after rooftop survey
Local Hardware Source	Physical / Local	Same day	VERIFY	Radiation shield — fabricate locally or source equivalent. Quotation required before final budget confirmation.

1.1 Procurement Controls

- Capture a dated quotation or checkout screenshot for every item before approval is submitted.
- Confirm model, quantity, dimensions, connector type, warranty terms, and delivery charge for each component before placing an order.
- For all Daraz.pk listings: verify seller history, review recency, return policy, and item authenticity before purchasing.
- Bench-test all electronic modules (sensors, ESP32, SD module) before committing to rooftop installation.
- Reconfirm all prices at point of order — listings are subject to change without notice.

2. Complete Line-Item Pricing

2.1 Core Sensing, Control & Protection Build

The following 13 components constitute the minimum viable sensor assembly required to begin data collection. Item 11 (Radiation Shield) is excluded from the subtotal pending a local quotation.

#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
1	PM Sensor	PMS7003 PM1/PM2.5/PM10 Laser Sensor	Embedded Studio	1	4,200	4,200
2	Microcontroller	ESP32 WROOM-32D Development Board	Digilog.pk	1	1,160	1,160
3	Environmental Sensor	BME280 Temperature / Humidity / Pressure	Electrobes	1	850	850

4	Data Logging Module	Arduino MicroSD Card Reader Module	Electrobes	1	150	150
5	Storage Media	Samsung EVO Plus 16 GB MicroSD Card	Daraz.pk	1	1,450	1,450
6	Weather Enclosure	Waterproof Electrical Enclosure, IP65-rated	A.E Solution	1	750	750
7	Connectivity Wiring	20 cm Mixed Jumper Wire Kit, 120 pcs	Daraz.pk	1	440	440
8	Power Supply	5V 2A AC/DC Power Adapter	Electronics Hub	1	170	170
9	Mounting — Cable Ties	Black Nylon Self-Locking Cable Ties	Daraz.pk	1	120	120
10	Mounting — Clamps	2-inch Stainless-Steel Hose Clamps, 2 pcs	DreamsMart.pk	1	60	60
11	Radiation Shield	Solar Radiation Shield / DIY Stevenson Screen	Local source required	0	TBD	TBD
12	Rain Proxy Sensor A	Raindrop Detection Sensor Module	Digilog.pk	1	160	160
13	Rain Proxy Sensor B	Rain Drop Moisture Detection Sensor Module	Electrobes	1	150	150
CORE BUILD PRICED SUBTOTAL (Items 1–10, 12–13 Item 11 excluded — local quote pending)						PKR 9,660

Item 11 is a reference specification only. Source or fabricate a white, ventilated radiation shield locally that blocks direct solar loading while maintaining ambient airflow. The Misol product listed in Appendix A is a dimensional reference — not a procurement source. Obtain a local quotation and add to the final approved total before procurement.

2.2 Validation & Deployment Additions

The following four items are additions to the core build that are required for a complete, robust rooftop deployment. The UNI-T anemometer provides periodic wind validation against public API data; the extension cable routes power to the rooftop; the sealant weatherproofs cable penetrations; and the fasteners secure the mounting assembly.

#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
14	Validation Anemometer	UNI-T UT363 Digital Anemometer	Electrobes	1	4,250	4,250
15	Power Extension Cable	Clopal 10 m Heavy-Duty, 5-Way Extension Lead	Clopal Online	1	3,295	3,295
16	Weatherproof Sealant	GMSA RTV Silicone Caulk Sealant, 310 ml	Expert Tools World	1	390	390
17	Mounting Fasteners	M3–M6 Mild-Steel Nuts and Bolts (8 pcs)	Multan Electronics	8	24	192
VALIDATION & DEPLOYMENT ADDITIONS SUBTOTAL (Items 14–17)						PKR 8,127
BASELINE HARDWARE SUBTOTAL (Core + Additions, Items 1–10, 12–17 Excl. Item 11)						PKR 17,787

2.3 Extension-Cable Alternative Scenario

The Camelion CMS-178 extension reel is an alternative to the baseline Clopal lead, not an additional item. If rooftop cable management or rating requirements favour a reel format, substitute Item 18 for Item 15. The cost difference is PKR 905 before contingency.

#	Component / Item	Model / Specification	Vendor	Qty	Unit (PKR)	Total (PKR)
18	Power Extension Alternative	Camelion CMS-178 Extension Reel, 10 m	Powerhouse Express	1	4,200	4,200

Substituting Item 18 for Item 15 increases the alternative hardware subtotal to PKR 18,692 (+PKR 905) and the alternative total budget (including 5% contingency) to PKR 19,627. Use the Clopal option (Item 15) for baseline approval.

3. Final Costing Statement

Cost Statement	PKR	Basis
Baseline priced hardware	17,787	Items 1–10 and 12–17 at listed prices. Item 11 (radiation shield) excluded pending local quotation.
Contingency reserve ~ 5%	889	Rounded to nearest rupee. Covers minor price movements at time of order.
TOTAL BASELINE BUDGET REQUEST	PKR 18,676	Recommended approval amount. Radiation shield to be added after local quotation.
Alternative hardware subtotal (Item 18 replaces Item 15)	18,692	Camelion reel substituted for Clopal lead. PKR 905 increase before contingency.
Alternative contingency ~ 5%	1,308	Rounded to nearest rupee.
TOTAL WITH CAMELION ALTERNATIVE	PKR 20,000	Alternative scenario only. Excludes radiation-shield local quote.

3.1 Exclusions & Assumptions

- Radiation shield (Item 11): excluded from both totals. Add local quotation to final approved amount before procurement.
- Delivery charges, payment fees, and applicable taxes are excluded unless already embedded in the listed vendor price.
- Fastener costing (Item 17) assumes eight units at PKR 24 each. Final quantity and size should be confirmed after the rooftop pre-installation survey.
- Only one extension-cable option is included in each scenario. Item 18 replaces — not supplements — Item 15.
- All prices are subject to vendor stock levels and checkout confirmation at point of order.

4. Component Procurement Notes

Critical technical and sourcing guidance for each component is set out below. These notes should be reviewed before placing orders and shared with the team member responsible for bench testing.

#	Component	Critical Procurement Note	Vendor Source
1	PMS7003 PM Sensor	Confirm the exact PMS7003 model and UART interface. Do not substitute any other Plantower variant without verifying frame format, dimensions, and firmware compatibility. The sensor must have an engineered air path through the enclosure — do not seal it in stagnant air.	Embedded Studio →
2	ESP32 WROOM-32D	Confirm Wi-Fi (802.11 b/g/n) is active and that the board exposes UART, I2C, and SPI pins. Bench-test USB programming and power stability before integrating with other modules.	Digilog.pk →
3	BME280 Environmental Sensor	Confirm BME280 — NOT BMP280. Humidity measurement is required for the ML feature set. Verify chip marking and cross-check temperature, humidity, and pressure readings during bench testing before installation.	Electrobes →
4	MicroSD Module	Verify SPI compatibility with ESP32. Confirm the module accepts 16GB cards. Test write speed and data integrity before outdoor deployment.	Electrobes →
5	Samsung EVO Plus MicroSD	For Daraz purchases, verify seller rating (minimum 4.5/5), recent reviews, and return policy before paying. Confirm 16GB capacity, FAT32 compatibility, and Samsung packaging on receipt.	Daraz.pk →
6	IP65 Enclosure	Confirm dimensions accommodate all components. Verify gasket condition, cable-entry arrangement, and wall-mounting footprint. The PMS7003 requires an engineered air path — the enclosure must not seal the sensor in stagnant air.	A.E Solution →
11	Radiation Shield (TBD)	The Misol product is a dimensional and functional reference only. Source or fabricate a white, ventilated shield locally that blocks direct solar loading while maintaining ambient airflow around the BME280. Obtain local quotation and submit for supplementary approval.	Reference spec →
12 & 13	Dual Rain Sensors	Purchase both modules as specified. Label them A and B. Compare analog and digital responses during bench testing and select the more stable unit for field deployment. Retain the other as a validation spare.	Digilog.pk (A) → Electrobes (B) →

14	UNI-T UT363 Anemometer	Use for periodic spot checks against public API wind data. For each comparison, record time, location, unit, averaging method, and weather conditions. Not continuously deployed — handheld spot-check instrument only.	Electrobes →
15 / 18	Extension Cable	Use the Clopal 10 m lead as the baseline option. If rooftop conditions require a reel format, enhanced cable management, or a higher-rated product, substitute the Camelion CMS-178 (Item 18) and use the alternative total in Section 3.	Clopal (baseline) → Powerhouse Express (alt.) →
16	Weatherproof Sealant	Apply sealant only after wiring and bench testing are fully complete. Maintain serviceability of the enclosure — do not permanently seal access points. Do not block sensor ventilation openings.	Expert Tools World →
17	Mounting Fasteners	Confirm clamp diameter, bolt size, corrosion resistance, and bracket geometry during the pre-deployment rooftop survey. Replace the provisional quantity of 8 units with the final bill of materials before purchasing.	Multan Electronics →

5. Procurement Timeline

The following timeline applies from the day budget approval is received. All timeline milestones are contingent on vendor stock availability and delivery performance at time of order.

Day	Activity	Vendor / Location	Control Note
Day 1	Place primary electronics orders: PM sensor, ESP32, BME280, SD module, rain sensors	Embeded Studio, Digilog.pk, Electrobes	Confirm stock, model specification, and delivery estimate before completing payment. Capture order confirmations.
Day 1	Place enclosure, power supply, and marketplace orders: IP65 box, power adapter, storage media, wiring, ties, clamps	A.E Solution, Electronics Hub, Daraz.pk, DreamsMart.pk	Verify Daraz seller ratings before purchase. Capture order screenshots with timestamps.
Day 1–2	Order extension cable, sealant, and anemometer. Confirm fastener specifications.	Clopal Online / Powerhouse Express, Expert Tools World, Multan Electronics	Confirm cable length, connector type, and sealant quantity. Confirm fastener sizes against pre-survey measurements.
Day 2	Obtain local radiation shield quotation	Local fabricator / hardware source (Islamabad)	Use Misol product dimensions as reference specification. Submit quotation for supplementary budget approval. Do not proceed to purchase without approval.

Day 3–7	Receive, inspect, and inventory all delivered items	Project team — campus	Photograph packaging and verify model/part numbers on receipt. Log any discrepancies immediately with the relevant vendor.
Day 6–8	Bench assembly and full component validation	Campus lab or workspace	Test power stability, sensor communications (I2C/SPI/UART), SD card logging, and Wi-Fi transmission. Do not proceed to rooftop installation until all modules pass bench validation.
Day 9	Rooftop installation and live system commissioning	BIC campus rooftop	Verify mounting security, enclosure sealing, airflow to sensors, Wi-Fi signal strength, and live data flow to campus workstation. Confirm SD card logging is active.

6. Approval & Procurement Record

Role	Name	Signature	Date
Prepared / Checked by			
Technical Approval			
Budget Approval			

All prices listed in this report are supplier-listed figures as of 26 July 2026 and must be reconfirmed at point of order. The radiation shield (Item 11) quotation must be obtained and submitted for approval before the final procurement order is placed.

Appendix A. Vendor Source Directory

All clickable source links below correspond to the supplier listings used for this updated pricing report. Each URL should be opened and the listing rechecked immediately before procurement to verify stock, price, and specification.

#	Component	Vendor	Source URL
1	PM Sensor	Embedded Studio	https://embededstudio.com/products/pms7003-dust-laser-sensor-module-embeded-studio
2	Microcontroller	Digilog.pk	https://digilog.pk/products/espressif-esp32-wroom-32d-development-board
3	Environmental Sensor	Electrobes	https://electrobes.com/product/bme280-altimeter-barometric-pressure-temperature-humidity-sensor-module-3-3v/
4	Data Logging Module	Electrobes	https://electrobes.com/product/micro-sd-tf-card-reader-adapter-module-for-arduino/
5	Storage Media	Daraz.pk	https://www.daraz.pk/products/16-10-i1948951637-s14012512909.html
6	Weather Enclosure	A.E Solution	https://www.aesolutions.com.pk/products/ip65-power-distribution-box-waterproof-electrical-enclosure
7	Connectivity Wiring	Daraz.pk	https://www.daraz.pk/products/120-20-i206016876-s4008826029.html
8	Power Supply	Electronics Hub	https://electronicshub.pk/product/5v-2a-power-supply-ac-dc-adapter/
9	Mounting — Cable Ties	Daraz.pk	https://www.daraz.pk/products/best-quality-black-nylon-cable-tie...
10	Mounting — Clamps	DreamsMart.pk	https://www.dreamsmart.pk/product/dreams-2pcs-clamp-2-inches...
11	Radiation Shield (Ref.)	Local source — TBD	Misol reference spec (dimension guide only) →
12	Rain Sensor A	Digilog.pk	https://digilog.pk/products/rain-drops-detection-sensor-module-soil-moisture-sensor-module-in-pakistan
13	Rain Sensor B	Electrobes	https://electrobes.com/product/rain-drop-moisture-detection-sensor/
14	Validation Anemometer	Electrobes	https://electrobes.com/product/uni-t-ut363-wind-speed-and-temperature-meter-digital-anemometer/
15	Power Extension (Baseline)	Clopal Online	https://clopal.com/products/clopal-extension-lead-with-10-mtrs-heavy-duty-copper-cable-5-ways-power-sockets-extension-board-with-wire
16	Weatherproof Sealant	Expert Tools World	https://experttoolsworld.com/product/gmsa-rtv-silicone-caulk-sealant-bottle-310ml/
17	Mounting Fasteners	Multan Electronics	https://multanelectronics.com/shop/m3-m4-m5-m6-mild-steel-nut-bolts-screw-round-cross-bolt-half-round/
18	Power Extension (Alternative)	Powerhouse Express	https://powerhouseexpress.com.pk/products/camelion-extension-reel-cms-178

AirSense Pakistan | Confidential — For Procurement Review Only | Price Review Date: 26 July 2026